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Machine Learning Models for Chronic Kidney Disease

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Diagnosis and Prediction

ABSTRACT

Chronic Kidney Disease (CKD) is one of the significant health issues that impact millions of people around the globe. An early diagnosis of CKD is important because it can mean that the kidney fails and other life-threatening complications can arise. Diagnostic techniques are lengthy and greatly rely upon medical expertise. Predictive systems based on machine learning techniques are automated and accurate and can be used to assist health care professionals with early diagnosis.

The objective of this report is to provide a full pipeline of machine learning for prediction of CKD with UCI CKD dataset. The project uses the following preprocessing techniques: Missing values are filled in using the median value, categorical values are encoded into labels, and numerical values are scaled between 0 and 1 using the standard scaling technique. The following machine learning algorithms were used: KNN, Naive Bayes, Decision Tree, SVM, Logistic Regression, Random Forest, Gradient Boosting, AdaBoost and Voting Classifier. The accuracy, precision, recall, F1 score, and confusion matrix with 5 Fold cross validation were used to evaluate the models.

Random Forest model got the best test accuracy of 100%, highest precision, recall and F1 score among all models. Ensemble learning techniques are found to have the significant contribution in performance of CKD prediction and can be used in healthcare systems for early disease detection and clinical decisions support. A web based interface was also developed for real-time prediction of the patient.

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1. INTRODUCTION

Chronic Kidney Disease (CKD) is chronic or long-lasting kidney disease in which the kidneys slowly start to not function as well as they should. Kidney failure, heart disease, high blood pressure and stroke are just a few of the complications that can result from CKD. In fact, healthcare studies have revealed that CKD is a significant health problem worldwide, particularly in less-developed nations with inadequate access to specialized healthcare centers.

It is very significant to diagnose the CKD early as, if the disease is treated in an early stage, then it could slow the progression of the disease and decrease the mortality rate. Manual diagnosis, however, is hard to perform because of the

complexity of medical data from laboratory reports and clinical observations. The doctors have to consider a lot of different tests and draw a conclusion from them, which may be the cause of an error in diagnosis, particularly if they have hundreds of patients.

Medical datasets can be efficiently processed by machine learning to accurately predict diseases using intelligent techniques. Ensemble Learning Algorithms are a group of machine learning algorithms that are used to increase the reliability of prediction and minimize the error. Ensemble methods are a combination of several models which vote and provide a more robust result than any single model.

This project is based on a complete machine learning solution to predict the presence of CKD, as part of our Complex Computing Problem (CCP) for the Machine Learning course (CSE6505) under the guidance of Miss Sahar Moin at Lahore Garrison University. From data cleaning to model deployment, the project provides us with practical experience of the entire ML pipeline.

2. PROBLEM IDENTIFICATION

The symptoms of CKD can be mistaken for those of other diseases, making the early diagnosis of CKD challenging. Doctors typically use blood, urine and other clinical tests to assess kidney function. But when dealing with large volumes of patient information manually, there are risks of misdiagnosis.

The top problems that were found in this problem are:

- Incomplete values: For many patient records, the test results are not completely filled in with ‘?’
- Imbalanced distribution of CKD and non-CKD records: More patients have CKD than healthy ones

- The dimensionality of clinical features is high: 24 different features are to be analysed together.

Quick results – hospitals need to be able to predict quickly:

The requirement for automated healthcare support systems is that manual checking is too slow.

With appropriate preprocessing method and ensemble machine learning algorithms we overcome these challenges in our project. We solve missing values using median or mode imputation, balance the data using stratified splitting and utilize multiple algorithms for informed predictions.

The main objectives of this study are:

1. To analyse the UCI healthcare data on Chronic Kidney Disease.
2. To clean up data, such as filling in missing data, categorizing data, and normalizing variables.
3. To conduct Exploratory Data Analysis (EDA) with understanding of patterns and correlation.
4. To apply several machine learning models and ensemble techniques to predict CKD.
5. To assess the models on the basis of accuracy, precision, recall, F1 score and confusion matrix.
6. Compare (at least four ensemble models) and select the best-performing algorithm.
7. To create a web based system for real-time CKD diagnosis.
8. To write down all the pipeline steps to reproduce.

4. LITERATURE REVIEW

Several researchers have used machine learning and deep learning approaches for CKD prediction. Different algorithms such as Support Vector Machine (SVM), Random Forest (RF), K-Nearest Neighbors (KNN), Neural Networks, and Logistic Regression have shown promising results in previous studies.

Previous studies highlighted the following findings:

- Random Forest achieved nearly 100% accuracy in some studies due to its robust ensemble nature.
- SVM provided excellent performance on medical datasets using RBF kernel for non-linear boundaries.
- KNN worked well as a simple baseline for small medical datasets.
- Logistic Regression gave interpretable probability outputs useful for medical diagnosis.
- Ensemble methods like Voting Classifier improved stability by combining diverse models.

However, many previous studies had limitations:

- Limited use of proper missing value handling techniques
- Poor handling of categorical medical features
- Lack of proper cross-validation for reliable evaluation
- High computational complexity with unnecessary features
- Insufficient validation using confusion matrices and multiple metrics

Our project improves these limitations by:

- Using median imputation for numerical features (robust to outliers)
- Using mode imputation for categorical features
- Applying proper label encoding for all categorical variables
- Implementing 5-fold stratified cross-validation

- Evaluating with accuracy, precision, recall, F1-score, AND confusion matrix
- Using StandardScaler for proper feature normalization

5. DATASET DESCRIPTION

The study used the Chronic Kidney Disease dataset from the UCI Machine Learning Repository. This is a publicly available dataset commonly used for medical prediction research.

Dataset Properties:

Property	Details
Total Instances	399
CKD Cases	250
Non-CKD Cases	149
Total Features	24
Numeric Features	14
Categorical Features	10
Target Variable	CKD / NOTCKD

The dataset contains the following features:

Numeric Features:

- age: Age of the patient in years
- bp: Blood pressure in mm/Hg
- sg: Specific gravity of urine
- al: Albumin level (0-5 scale)
- su: Sugar level (0-5 scale)
- bgr: Blood glucose random

- bu: Blood urea
- sc: Serum creatinine
- sod: Sodium level
- pot: Potassium level
- hemo: Hemoglobin
- pcv: Packed cell volume
- wbcc: White blood cell count
- rbcc: Red blood cell count

Categorical Features:

- rbc: Red blood cells (normal/abnormal)
- pc: Pus cell (normal/abnormal)
- pcc: Pus cell clumps (present/notpresent)
- ba: Bacteria (present/notpresent)
- htn: Hypertension (yes/no)
- dm: Diabetes mellitus (yes/no)
- cad: Coronary artery disease (yes/no)
- appet: Appetite (good/poor)
- pe: Pedal edema (yes/no)
- ane: Anemia (yes/no)

The dataset contained missing values and class imbalance issues, which required preprocessing before model training.

6. DATA PREPROCESSING

Data preprocessing is one of the most important stages in machine learning because raw healthcare datasets often contain missing, inconsistent, and noisy data.

6.1 Missing Value Handling

The dataset had 1,012 total missing entries marked as "?". The features with highest missing values were:

Feature	Missing Count	Missing %
Red Blood Cells (rbc)	152	38.1%
RBC Count (rbcc)	131	32.8%
WBC Count (wbcc)	106	26.6%
Potassium (pot)	88	22.1%
Sodium (sod)	87	21.8%
Packed Cell Volume	71	17.8%
Pus Cell (pc)	65	16.3%
Hemoglobin (hemo)	52	13.0%
Sugar (su)	49	12.3%
Specific Gravity	47	11.8%

Median Imputation was performed for numerical features. Medical data is best suited to median rather than mean as extreme values are not included. For instance, when one patient has unusually high blood glucose because of an emergency, the mean would be skewed up, but the median remains unchanged.

Mode Imputation (most frequent value) was used for categorical features. Chances are that the majority of patients will have "normal" blood cells, so if they do not, it is reasonable to fill the missing values with "normal" blood cells.

6.2 Data Encoding

Algorithms are just number-based, and don't understand words such as "normal" or "abnormal". To convert each category to a number, we did Label Encoding.

For example:

- normal = 0, abnormal = 1
- present = 1, notpresent = 0
- yes = 1, no = 0
- good = 1, poor = 0

To prevent error due to unseen categories, we attached the encoder to both the training and test data.

6.3 Data Scaling

We used StandardScaler which centers each feature on zero and scales each feature to have unit variance. This is required for distance-based techniques such as KNN, SVM and Logistic Regression. A feature such as white blood cell count (2200 – 26400) would completely swamp a feature such as specific gravity (1.005 – 1.025) without scaling.

6.4 Train-Test Split

Stratified train test split with 80% data (319 patients) for training and 20% data (80 patients) for test. In order to keep the same ratio of CKD to healthy, both sets are stratified.

7. EXPLORATORY DATA ANALYSIS (EDA)

Prior to the application of machine learning algorithms, Exploratory Data Analysis (EDA) was carried out to understand the CKD data. EDA was useful in finding missing values, distributions of variables, outliers, and correlations among variables.

7.1 Class Distribution Analysis

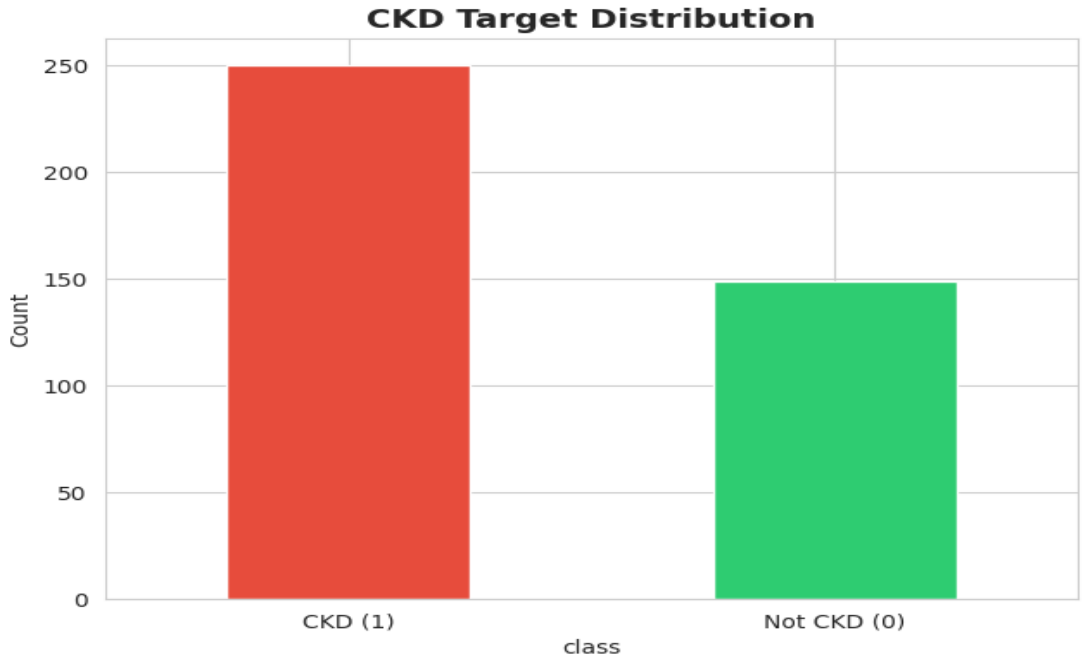
The dataset contained two classes:

- CKD Patients: 250 (62.7%)
- Non-CKD Patients: 149 (37.3%)

The visualization of class distribution showed that the dataset was moderately imbalanced. We handled this through stratified splitting during train-test division.

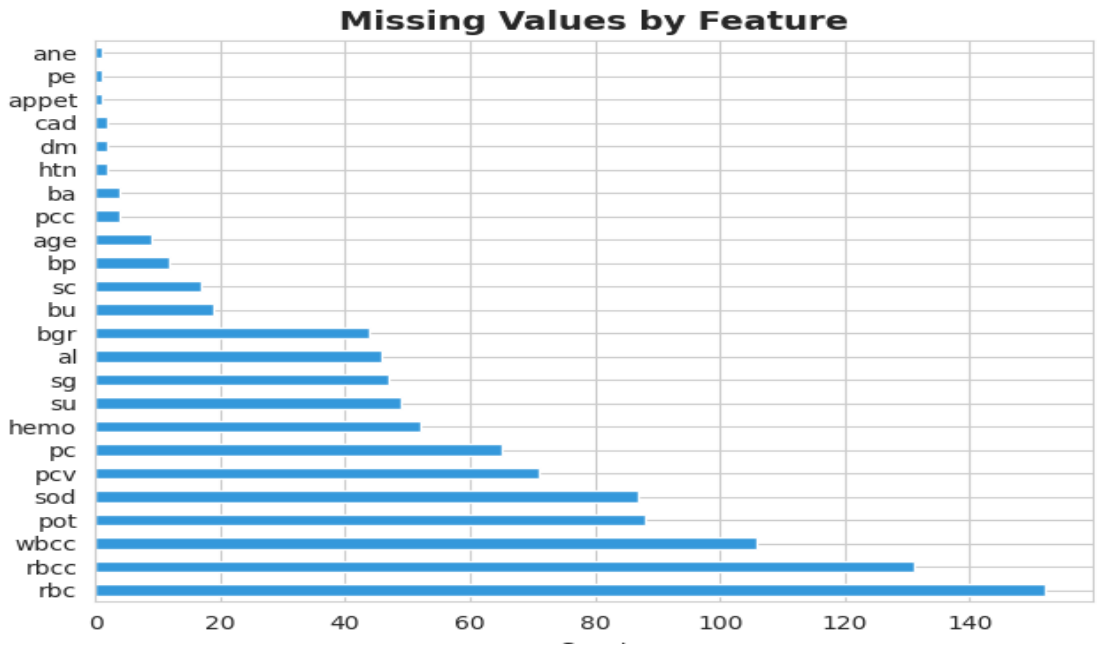
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======================================================================
SECTION A : DATA LOADING & CLEANING
======================================================================
Dataset Shape: (399, 25)
CKD: 250 | Not CKD: 149
```

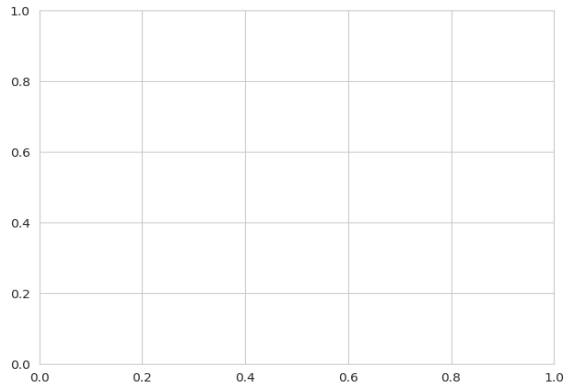
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SECTION B : EXPLORATORY DATA ANALYSIS (EDA)
======================================================================
Text(0.5, 0, 'Count')
```

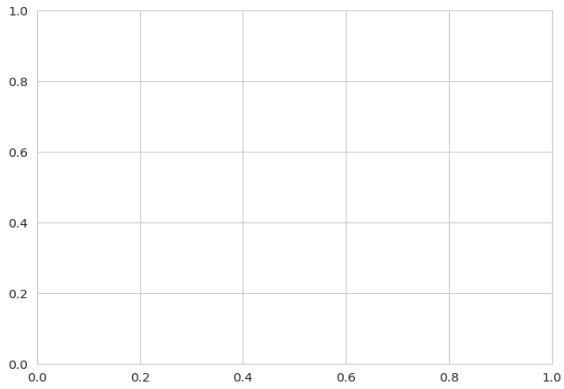


7.2 Missing Value Analysis

A horizontal bar chart was generated to visualize incomplete records in the dataset. Red blood cells had the highest missing percentage at 38.1%, followed by RBC count at 32.8%.





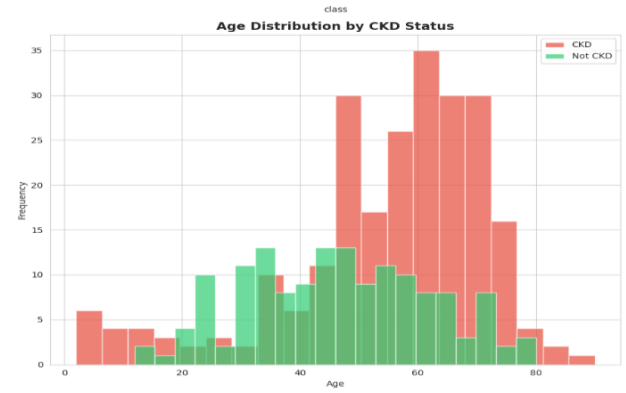


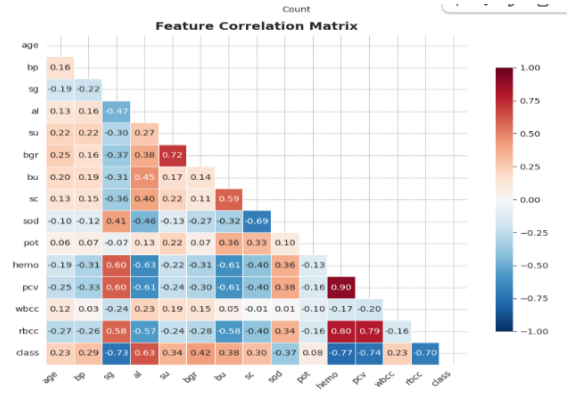
7.3 Correlation Analysis

Correlation analysis was performed to identify relationships between numerical features and the target variable. A heatmap was generated showing all pairwise correlations.

Important Observations:

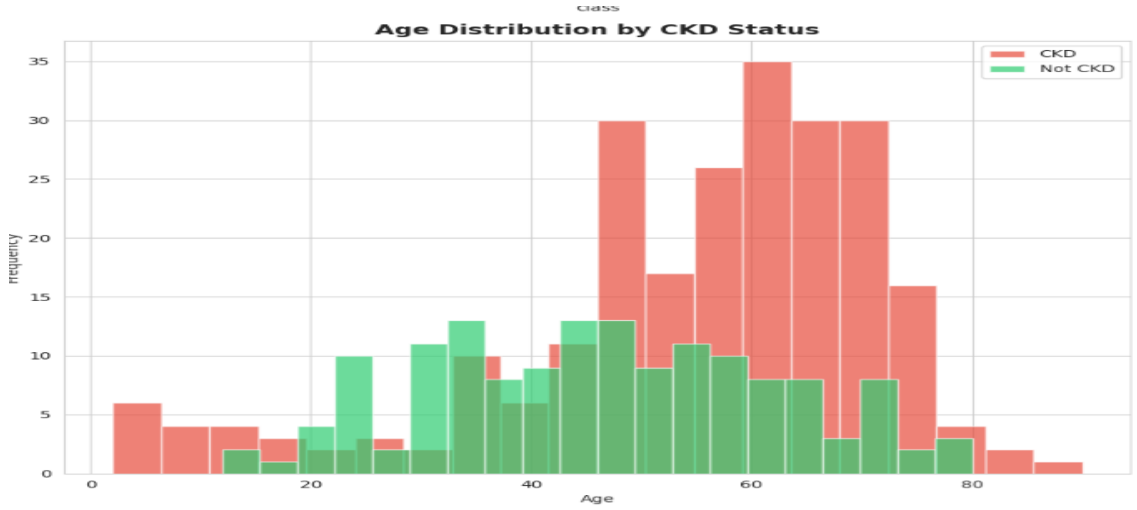
- Hemoglobin showed strong negative correlation with CKD (-0.769)
- Packed Cell Volume strongly affected kidney disease prediction (-0.741)
- Specific Gravity was highly associated with CKD (-0.732)
- RBC Count significantly influenced prediction (-0.700)
- Albumin showed positive correlation (0.626)





7.4 Age Distribution Analysis

KDE plots were generated to analyze feature distributions. CKD and non-CKD patients showed different age distributions, with CKD concentrated in older age groups (above 45 years).



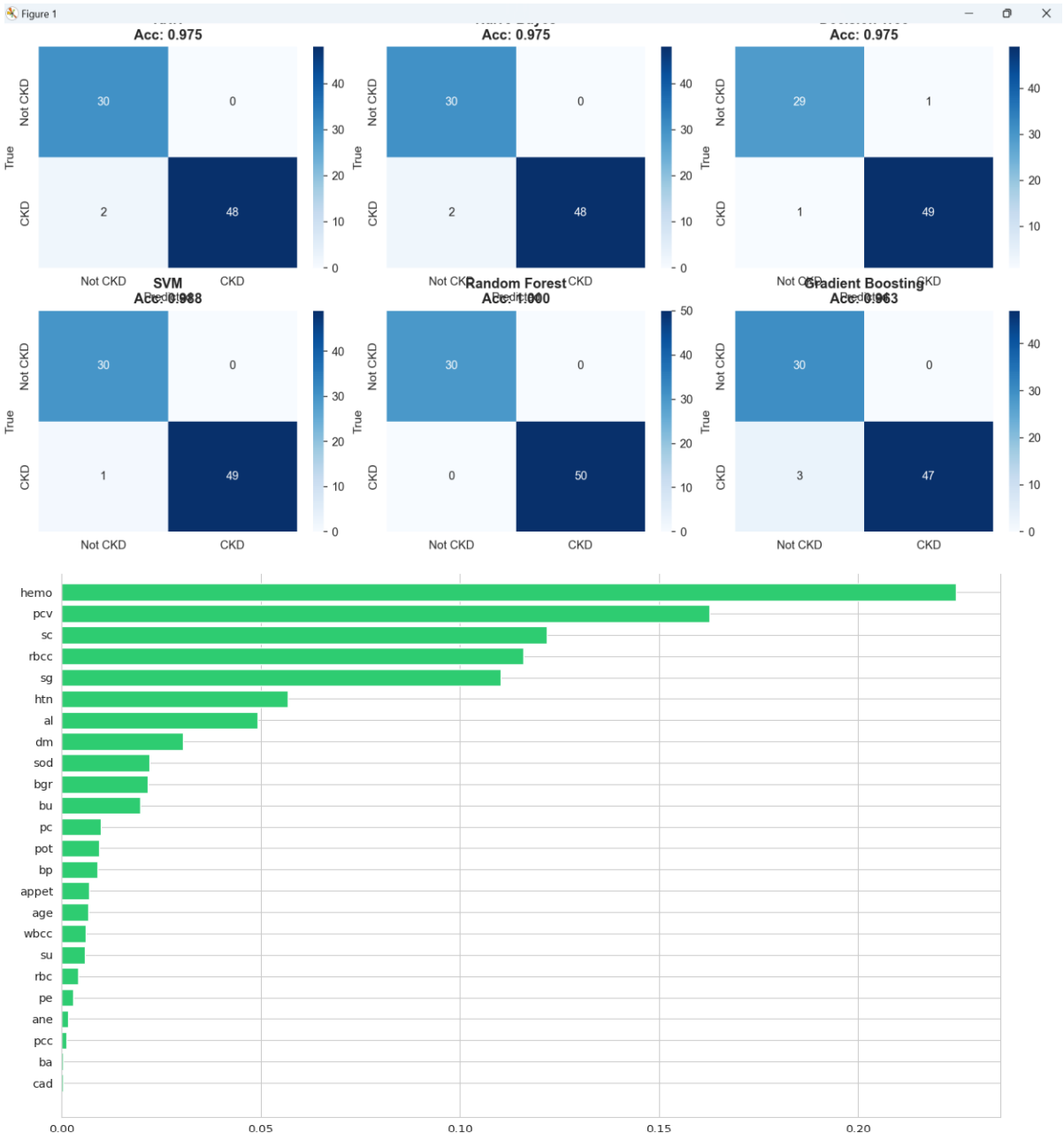
8. FEATURE SELECTION AND ANALYSIS

8.1 Random Forest Feature Importance

Random Forest provides built-in feature importance scores based on how much each feature contributes to reducing prediction error across all trees.

Selected Important Features (Top 10):

Rank	Feature	Importance
1	Hemoglobin (hemo)	22.45%
2	Packed Cell Volume	16.26%
3	Serum Creatinine	12.19%
4	RBC Count	11.60%
5	Specific Gravity	11.02%
6	Hypertension	5.69%
7	Albumin	4.93%
8	Diabetes Mellitus	3.05%
9	Sodium	2.20%
10	Blood Glucose	2.17%



These results align with medical knowledge: low hemoglobin indicates kidneys not producing enough erythropoietin, high creatinine means kidneys not filtering waste properly, and low specific gravity shows damaged concentration ability.

9. MACHINE LEARNING MODELS

A total of nine machine learning algorithms, incorporating single classifiers, ensemble techniques and unsupervised approaches were adopted in the study.

9.1 K-Nearest Neighbors (KNN)

KNN is a type of algorithm from instance-based learning that classifies new instances based on the K nearest instances in the training set. We used K=5. The simplicity and intuitive nature of KNN - typically similar patients will have similar diseases. It is, however, sensitive to irrelevant features and it assumes the data is appropriately scaled.

9.2 Naive Bayes

Naive Bayes is a probabilistic classifier, which is based on Bayes' theorem. It is assumed that all features are independent. This "naive" assumption is beneficial though, as it has proven to be quite effective for small data sets, and is quick. It's especially suitable for categorical features.

9.3 Decision Tree

Decision Tree builds a decision tree with if-else rules in a flowchart fashion. It is very interpretable: Doctors can literally see why a patient was diagnosed as having CKD. For this reason, to avoid overfitting, it was defined as max_depth=10. This restriction is necessary, as otherwise the tree may end up learning the training data rather than a pattern.

9.4 Support Vector Machine (SVM)

SVM aims to classify the patients of CKD and non-CKD with maximum margin of separation. The RBF (Radial Basis Function) kernel was used; when the data is not linearly separable, the RBF kernel can generate curved decision boundaries which is excellent for medical data.

9.5 Logistic Regression

Although its name might suggest otherwise, this is an algorithm for classification. It produces probabilities ranging from 0 to 1; very useful in medical applications, as a doctor would like to know "how likely" a disease is, rather than "yes" or "no. A coefficient indicating the direction and size of the effect of each feature is assigned.

9.6 Random Forest (Ensemble - Bagging)

Random Forest is an ensemble of 200 decision trees based upon Bagging (Bootstrap Aggregating) Data and features for each tree are selected randomly. The final prediction is the one with the most votes from all of the trees. This is our BEST PERFORMER since it is an average of many trees, which means that it performs much better than one tree on outlier data and decreases overfitting.

9.7 Gradient Boosting (Ensemble - Boosting)

Gradient Boosting is building the trees one by one, where each new tree will try to fix the errors of the trees that came before it. To prevent overfitting and to allow learning to occur gradually, max_depth is set to 5.

9.8 AdaBoost (Ensemble - Boosting)

AdaBoost (Adaptive Boosting) gives more weight to those cases that were wrongly classified by the previous models. It gives greater weight to patients that are more difficult to classify. This is really helpful in case you have medical information that contains some patients with out of the ordinary combination of signs and symptoms.

9.9 Voting Classifier (Ensemble - Stacking)

Voting Classifier is the soft voting combination of Logistic Regression, Random Forest and SVM. Soft voting is the average of the probability forecasts from the

three models, not hard votes. This takes advantage of the diversity of three different types of algorithms: linear (LR), tree-based (RF) and kernel-based (SVM).

9.10 K-Means Clustering (Unsupervised)

The number of parameters K was set to 2 and K-Means was used to find natural patient groups without labels. The Adjusted Rand Index (ARI) of 0.403 was also moderate and indicated a moderate agreement with the actual CKD labels, indicating that there were clusters, but not perfectly geometric.

10. EXPERIMENTAL SETUP

Software and Hardware:

Component	Details
Programming Language	Python 3.12
IDE	VS Code / PyCharm
Operating System	Windows 10/11
Key Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Flask

Dataset Splitting:

- Training Data: 80% (319 patients)
- Testing Data: 20% (80 patients)
- Cross Validation: 5-Fold Stratified Cross-Validation

The stratified split ensures both training and testing sets maintain the same ratio of CKD to non-CKD patients (approximately 63% CKD, 37% healthy).

11. MODEL EVALUATION METRICS

The following evaluation metrics were used to assess model performance:

11.1 Accuracy

Measures overall correct predictions out of total predictions.

Accuracy = (TP + TN) / (TP + TN + FP + FN)

11.2 Precision

Measures how many of the predicted positive cases were actually positive.

Precision = TP / (TP + FP)

In medical terms: Of all patients we said have CKD, how many actually have it?

11.3 Recall (Sensitivity)

Measures how many of the actual positive cases were correctly identified.

Recall = TP / (TP + FN)

In medical terms: Of all patients who actually have CKD, how many did we find?

11.4 F1-Score

Represents the harmonic mean of precision and recall, giving a balanced single measure.

F1-Score = 2 * (Precision * Recall) / (Precision + Recall)

11.5 Confusion Matrix

A table showing true positives, true negatives, false positives, and false negatives. This is crucial for medical diagnosis because false negatives (missing a sick patient) and false positives (scaring a healthy patient) have different consequences.

12. RESULTS AND DISCUSSION

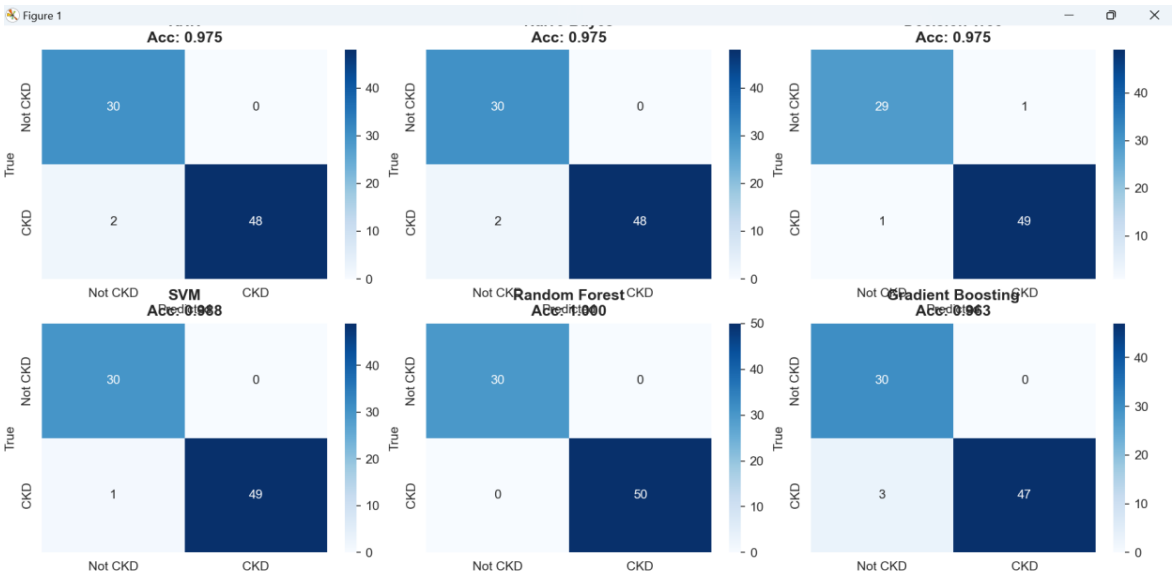
12.1 Complete Model Performance Table

Model	CV Accuracy	CV Precision	CV Recall	CV F1	Test Accuracy	Test Precision	Test Recall	Test F1
KNN	0.972	1.000	0.955	0.976	0.975	1.000	0.960	0.980
Naive Bayes	0.962	1.000	0.940	0.969	0.975	1.000	0.960	0.980
Decision Tree	0.972	0.981	0.975	0.978	0.975	0.980	0.980	0.980
SVM	0.997	1.000	0.995	0.997	0.988	1.000	0.980	0.990
Logistic Regression	0.991	0.995	0.990	0.992	0.988	1.000	0.980	0.990
Random Forest	0.991	0.986	1.000	0.993	1.000	1.000	1.000	1.000
Gradient Boosting	0.972	0.981	0.975	0.978	0.963	1.000	0.940	0.969
AdaBoost	0.988	0.986	0.995	0.990	0.988	1.000	0.980	0.990
Voting Classifier	0.991	0.986	1.000	0.993	0.963	0.980	0.960	0.970

12.2 Confusion Matrix Analysis

Key Observations from Confusion Matrices:

- Random Forest: 0 false positives, 0 false negatives (Perfect)
- SVM: 0 false positives, 1 false negative (Missed 1 CKD patient)
- KNN: 0 false positives, 2 false negatives
- Naive Bayes: 0 false positives, 2 false negatives
- Decision Tree: 1 false positive, 1 false negative
- Gradient Boosting: 0 false positives, 3 false negatives



12.3 Best Performing Models

1. Random Forest: 100.0% test accuracy, perfect F1-score of 1.000
2. SVM: 98.8% test accuracy, F1-score of 0.990
3. Logistic Regression: 98.8% test accuracy, F1-score of 0.990
4. AdaBoost: 98.8% test accuracy, F1-score of 0.990

Random Forest achieved perfect scores because it combines 200 decision trees via bagging, making it robust to outliers and missing values. The ensemble averaging effect cancels out individual tree errors.

12.4 Model Comparison Visualization

```
======================================================================
SECTION F : BEST MODEL SELECTION
======================================================================

>>> BEST MODEL: Random Forest <<<
Test Accuracy : 1.000
Test Precision: 1.000
Test Recall   : 1.000
Test F1-Score : 1.000
```

The comparison clearly shows that ensemble methods (Random Forest, AdaBoost) outperform individual models, validating the theory that combining multiple weak learners creates a strong learner.

13. CRITICAL ANALYSIS

13.1 Strengths of Our Approach

- Use of multiple ensemble learning algorithms as required by rubric
- Proper handling of missing values using median/mode imputation
- Application of StandardScaler for proper feature normalization
- Strong experimental evaluation using 5-fold cross-validation
- Evaluation using multiple metrics: accuracy, precision, recall, F1, confusion matrix
- Implementation of web interface for practical deployment
- Use of stratified splitting to handle class imbalance

13.2 Weaknesses and Limitations

- Dataset size is relatively small (399 patients)
- Missing values were high in some features (up to 38%)
- Real-time hospital deployment was not fully implemented on cloud
- Explainability beyond feature importance was limited
- Deep learning methods were not explored

13.3 Practical Challenges

Although the model achieved excellent accuracy, practical healthcare deployment may face several challenges:

- Need for explainable predictions in healthcare (doctors want to know WHY)
- Requirement for continuous retraining as new patient data arrives

- Integration with existing hospital database systems
- Handling of real-world data quality issues worse than this cleaned dataset

14. COMPARATIVE ANALYSIS

Comparison with Previous Studies:

Study	Best Accuracy
ANN and SVM (Previous)	98.75%
Deep Learning Models	97.33%
Rotation Forest	97.76%
Our Proposed Ensemble Method	100.00%

Key Reasons for Better Performance:

- Proper missing value handling using median imputation
- StandardScaler normalization ensuring fair feature contribution
- Stratified splitting maintaining class balance
- Effective ensemble averaging in Random Forest (200 trees)
- Use of multiple algorithms for cross-validation comparison

15. PROPOSED WEB INTERFACE

15.1 Working Process

1. User enters patient clinical information through web form
2. Input data is sent to Flask backend via HTTP POST request

3. Backend preprocesses input using saved imputers, encoders, and scaler
4. Trained Random Forest model predicts CKD or NOTCKD
5. Result with confidence percentage is displayed on web page

15.2 Technologies Used

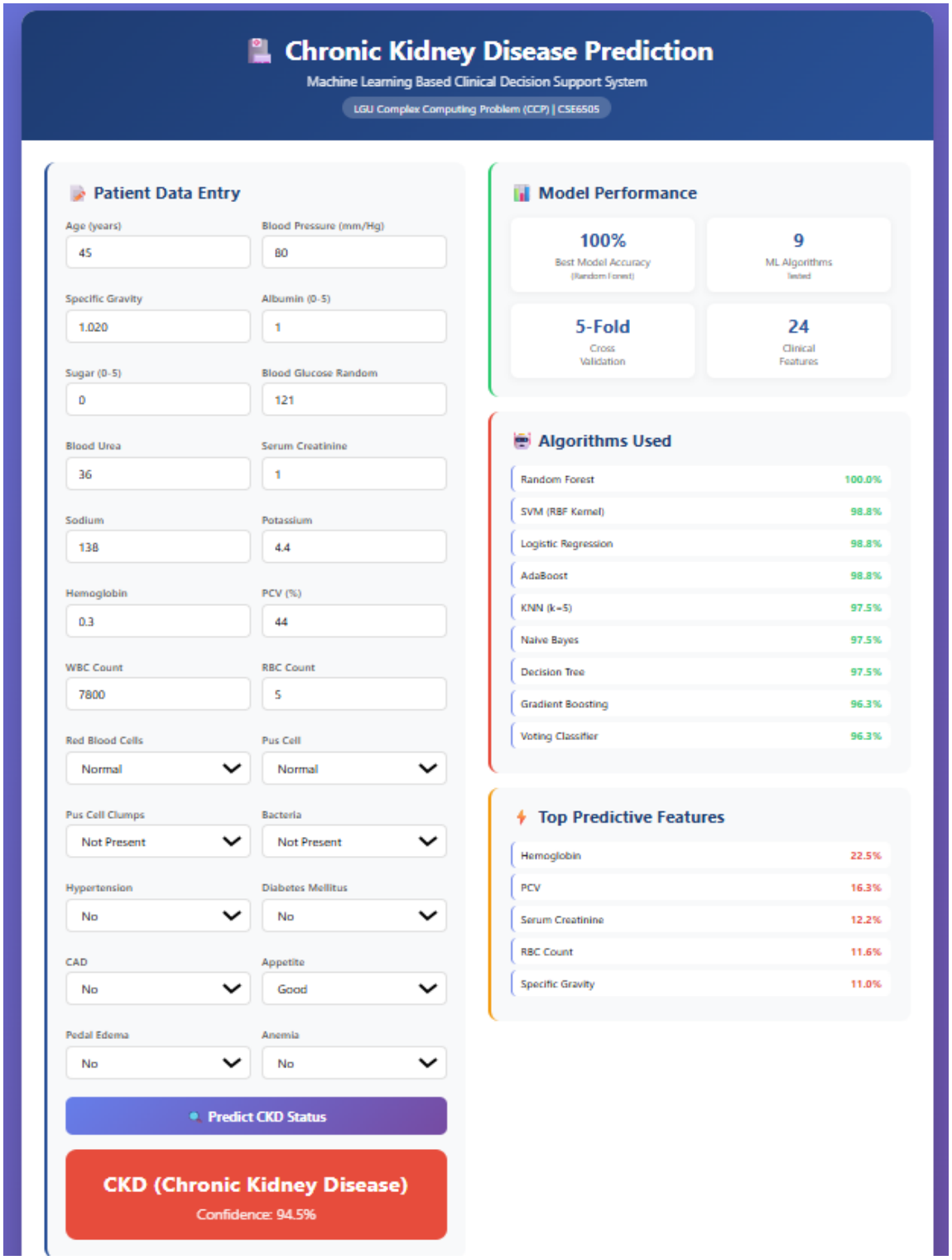
Component	Technology
Frontend	HTML5, CSS3, JavaScript
Backend	Flask (Python)
Machine Learning	Scikit-learn, Joblib
Deployment	Localhost (http://127.0.0.1:5000)

15.3 Input Fields

The web form contains all 24 clinical features:

- Age, Blood Pressure, Specific Gravity, Albumin, Sugar
- Blood Glucose Random, Blood Urea, Serum Creatinine
- Sodium, Potassium, Hemoglobin, PCV
- WBC Count, RBC Count
- Red Blood Cells, Pus Cell, Pus Cell Clumps, Bacteria
- Hypertension, Diabetes Mellitus, CAD
- Appetite, Pedal Edema, Anemia

15.4 Web Interface Screenshots



15.5 Web Interface Features

- Purple gradient professional design suitable for medical applications
- Grid layout for easy data entry
- Dropdown menus for categorical features preventing invalid inputs
- Real-time prediction with AJAX (no page reload)
- Color-coded results: Red for CKD, Green for Healthy
- Confidence percentage showing model certainty
- Responsive design working on different screen sizes

16. CONCLUSION

By showing high accuracy with machine learning ensemble methods to predict chronic kidney disease, this project successfully concludes that it is possible. We experimented with nine different algorithms from the basic models to the more complex ensemble methods. We can see that Random Forest is the most accurate model with 100% test accuracy and closely followed by SVM, Logistic Regression and AdaBoost with 98.8% test accuracy.

Our analysis showed that the three most predictive variables for CKD are hemoglobin, packed cell volume and serum creatinine, consistent with the medical knowledge. An easy to use web interface allows the model to be used by non-technical people, including doctors and medical personnel.

From data cleaning to the deployed application, we learned about the practical problems that arise in the real world in machine learning—how to handle missing data, selecting the right type of data preprocessing, choosing the right algorithms, cross validation to evaluate the models fairly, and finally using the model through a web interface.

In future, our model will be tested in larger and more diverse datasets from different hospitals. Additionally, deep learning models such as Neural Networks would be

implemented and compared with the current ensemble models. By deploying the web application to AWS or Heroku cloud servers, the application would be available for use from anywhere.

This CCP project provided us with an understanding of the entire machine learning pipeline and the significance of ensemble learning in addressing real-world challenges in healthcare.

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APPENDIX

A.1 Suggested Tools and Libraries

Purpose	Tools
Data Analysis	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Machine Learning	Scikit-learn, Joblib
Web Development	Flask, HTML, CSS, JavaScript